A reservoir-flow project, explained

What the baseline simulation did, in everyday language

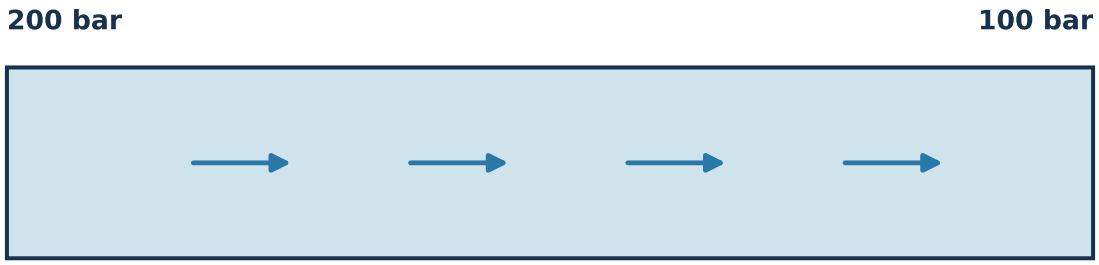
The big idea

This project estimates how fluid moves through underground rock. Imagine rock as a sponge: some rock lets water pass easily, while other rock resists it. The program calculates pressure inside the rock and the flow from a high-pressure side toward a low-pressure side.

The test case that ran

- A rectangular sample, 1,000 metres long and 200 metres tall.
- The rock was uniform: permeability (how easily fluid moves through it) was the same everywhere.
- Pressure was 200 bar on the left and 100 bar on the right. One bar is roughly atmospheric pressure.
- The top and bottom edges were sealed, so fluid could not leave through them.
- The computer divided the sample into 100 columns and 20 rows: 2,000 small cells.

A simple picture



Fluid moves from higher pressure toward lower pressure

This is a simplified teaching and testing case, not a prediction about a real reservoir.

How the computer worked it out

The solver uses a grid and balances flow cell by cell

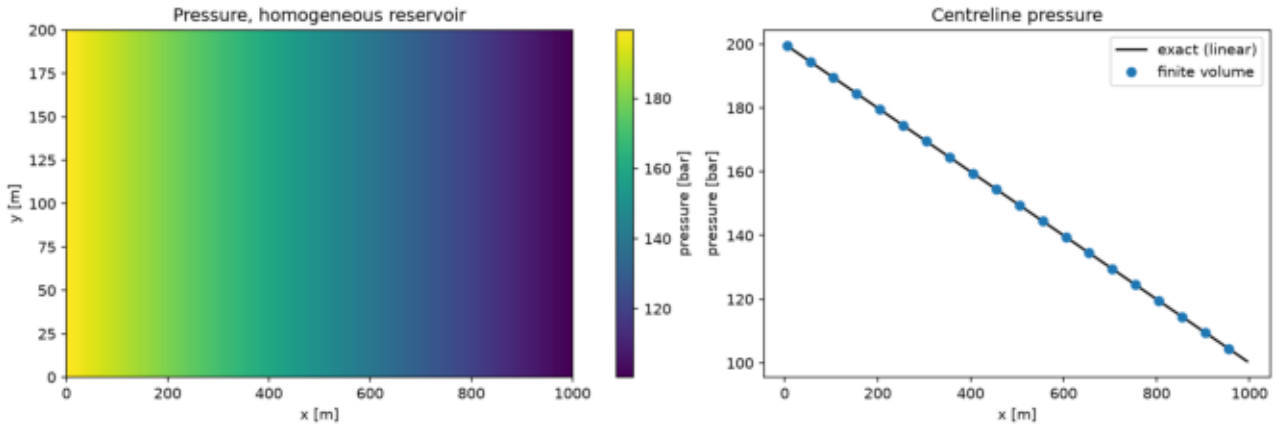
Think of each cell as a small box

The program stores a pressure in every box. It calculates how much fluid crosses each box edge, then solves for pressures that make inflow and outflow balance. This is called a finite-volume method: in everyday language, it carefully keeps track of what enters and leaves each box.

What permeability changes

Permeability measures how easily fluid passes through rock. Higher permeability means less resistance to flow. In this example permeability is the same everywhere, so pressure falls in a straight line from the left edge to the right edge.

The plot produced by the run



The plot shows pressure across the sample and compares computed centreline values with the exact straight-line answer.

Results and what I did

Five physics checks passed in the baseline run

The results

The computer compared its solution with physical rules and a known exact answer. The relative errors below are tiny; a relative error of zero means a perfect match for that check.

- Cell-by-cell balance: each cell’s incoming and outgoing flow matched. Relative error: 4.47×10^{-13}.
- Whole-sample balance: flow in from the left matched flow out on the right. Relative error: 9.12×10^{-12}.
- Pressure bounds: no cell went below 100 bar or above 200 bar. Error: 0.
- Exact straight-line answer: computed pressure matched the known uniform-rock solution. Error: 3.35×10^{-14}.
- Layered-rock reference formula: also matched. Error: 3.43×10^{-13}.

Headline numbers

- Computed flow rate: 0.000200 m^2/s per unit depth, the same as the exact formula.
- Maximum pressure difference from the exact answer: about 0.00000067 Pa, effectively zero for this test.
- Total: 5 out of 5 physics checks passed.

Work performed

- Installed the project’s declared packages in its existing .venv because required packages were missing.
- Ran examples/run_baseline.py; it created outputs/baseline.png.
- Did not change simulation source code or run the optional Ollama AI model.

Key files: examples/run_baseline.py (example), src/reservoir_agent/darcy_solver.py (solver), src/reservoir_agent/physics_validation.py (checks). This is a simplified test, not a real-field forecast.